Lecture 16

Complex exponentials and real polynomials

Uses.

Summary 2: Standing fact E supplies the real exponential, logarithm and trigonometric identities. Induction applies to finite degree reductions.
Summary 5: Real component limits preserve sums and products.
Summary 8: The real exponential and trigonometric derivative rules and real product and chain rules are proved.
Summary 11: Sinusoidal motions can be verified by substitution into a linearised oscillator equation.
Summary 15: Complex arithmetic is associative and distributive. Conjugation and modulus control division. Polar multiplication adds arguments.

Objectives. *Complex exponential notation records the polar multiplication law in one formula. We use complex roots to factor real polynomials into real linear and quadratic factors. Differentiation with respect to a real parameter then relates that notation to real oscillations and circular motion. The existence of complex polynomial roots is stated as a standing assumption.*

By the end of this lecture you will be able to:

- Prove the complex exponential product law and solve its elementary equations with all periods.
- Perform polynomial division and prove real factorisation using standing fact F.
- Differentiate the complex exponential with respect to a real parameter.
- Calculate sinusoidal amplitudes and phases and verify circular velocity and acceleration for positive radius and frequency.

16.1 Complex exponential notation

For a complex number $z = a + \mathrm{i}b$, define

$$e^z = e^a(\cos b + \mathrm{i}\sin b). \tag{16.1}$$

Here $a, b \in \mathbb{R}$ and e^a is the real exponential already used in calculus. For a purely imaginary number this gives $e^{\mathrm{i}b} = \operatorname{cis} b$. The new notation agrees with the old real exponential when $b = 0$. This is a definition through real functions. Lectures 31 and 34 identify it with a convergent complex power series, first on the real and imaginary axes and then for every complex argument.

Let $w = c + \mathrm{i}d$ be another complex number. The real exponential law and the polar multiplication law give

$$e^z e^w = e^{a+c}\operatorname{cis}(b+d) = e^{z+w}.$$

The modulus and conjugation properties are

$$|e^z| = e^{\operatorname{Re} z} > 0, \qquad \overline{e^z} = e^{\overline{z}}.$$

The first uses $|\operatorname{cis} b| = 1$. The second uses the evenness of cosine and oddness of sine. In particular, the complex exponential is never zero. Its periodicity follows from that of sine and cosine: $e^{z+2\pi k\mathrm{i}} = e^z$ for every $k \in \mathbb{Z}$.

For a nonzero value $w = R\operatorname{cis}\varphi$, all solutions of $e^z = w$ are

$$z = \ln R + \mathrm{i}(\varphi + 2\pi k), \qquad k \in \mathbb{Z}. \tag{16.2}$$

Indeed, taking moduli gives $e^{\operatorname{Re} z} = R$, hence $\operatorname{Re} z = \ln R$. The argument equation gives $\operatorname{Im} z = \varphi + 2\pi k$. Conversely, every displayed value has the required exponential by the definition. Thus the exponential has no single inverse on all of $\mathbb{C}$. For example, solving $e^z = -2 + 2\sqrt{3}\,\mathrm{i}$ gives

$$z = \ln 4 + \mathrm{i}(2\pi/3 + 2\pi k), \qquad k \in \mathbb{Z},$$

since the target has modulus 4 and argument $2\pi/3$. The equation $e^z = 0$ has no solution.

16.2 Polynomial division and multiplicity

A *complex polynomial* is a finite expression $P(z) = a_0 + a_1 z + \cdots + a_n z^n$ with complex coefficients. If $a_n \neq 0$, its *degree* is n and its *leading coefficient* is a_n. The zero polynomial has every coefficient zero; its degree is left undefined. A polynomial is *real* if all its coefficients are real. A *root* is a number $\alpha \in \mathbb{C}$ with $P(\alpha) = 0$. Two nonzero polynomials have a nonzero product whose degree is the sum of their degrees. This follows because the highest term is the product of their nonzero leading coefficients, and no other term has that degree.

Theorem 16.1 (Polynomial division). *For a polynomial P and a nonzero polynomial D of degree m, there exist unique polynomials Q and R with*

$$P = DQ + R,$$

where $R = 0$ or $\deg R < m$.

Proof. If D is a nonzero constant, take $Q = P/D$ and $R = 0$. Suppose $m \geq 1$. If $P = 0$ or its degree is below m, take $Q = 0$ and $R = P$. Otherwise let the leading terms be $a_n z^n$ for P and $d_m z^m$ for D. Subtract $(a_n/d_m)z^{n-m}D$ from P. The leading terms cancel, so the degree strictly decreases unless the new polynomial is zero. Repeat on the new polynomial while its degree is at least m. A decreasing list of nonnegative degrees must stop after finitely many steps. The sum of the subtracted multiples is DQ and the final polynomial is the required remainder R.

For uniqueness, two divisions would give $D(Q_1 - Q_2) = R_2 - R_1$. If $Q_1 - Q_2 \neq 0$, the left side has degree at least m. The right side is zero or has degree below m, a contradiction. Hence $Q_1 = Q_2$, and then $R_1 = R_2$. This also proves uniqueness when $m = 0$. ∎

Example 16.2 (A nonzero remainder)**.** Divide $P(z) = z^3 - 2z^2 + 3z + 1$ by $D(z) = z^2 + 1$. Subtracting $zD(z)$ leaves $-2z^2 + 2z + 1$. Subtracting $-2D(z)$ from this leaves $2z + 3$. The quotient and remainder are therefore

$$Q(z) = z - 2, \qquad R(z) = 2z + 3.$$

Their degrees satisfy $\deg R = 1 < 2 = \deg D$. Multiplication verifies $(z^2+1)(z-2)+(2z+3) = z^3 - 2z^2 + 3z + 1$.

A polynomial D *divides* P if $P = DQ$ for some polynomial Q. Dividing by $z - \alpha$ leaves a constant remainder. Evaluation at α shows that this remainder is $P(\alpha)$. Thus the *factor theorem* is

$$P(\alpha) = 0 \quad \Longleftrightarrow \quad z - \alpha \text{ divides } P.$$

For a nonzero polynomial, repeated division by this factor stops after at most its degree. The *multiplicity* of a root α is the positive integer k in the factorisation

$$P(z) = (z - \alpha)^k Q(z), \qquad Q(\alpha) \neq 0.$$

The stopping rule proves existence of such a largest power. Its exponent is unique: if a larger exponent also divided P, cancellation of the smaller power would make $Q(\alpha) = 0$. Cancellation is valid because a product of nonzero polynomials is nonzero.

A nonzero polynomial of degree n has at most n distinct roots. We prove this by induction on n. A nonzero constant has no roots. If a positive-degree polynomial has no root, the claim holds. If it has a root α, divide out $z - \alpha$. Every different root β satisfies $0 = (\beta - \alpha)Q(\beta)$ and hence $Q(\beta) = 0$. The quotient has degree $n - 1$, so the induction hypothesis gives at most $n - 1$ further distinct roots. This bound does not prove existence of any root.

16.3 Complex roots and real factors

Standing fact F: fundamental theorem of algebra. Every nonconstant complex polynomial has at least one complex root. We assume this existence theorem.

Combining F with the factor theorem gives a full factorisation of every nonzero degree-n complex polynomial:

$$P(z) = a_n \prod_{j=1}^{n} (z - \alpha_j).$$

Here the finite product multiplies the factors with indices $j = 1, \ldots, n$. Repeated roots occur repeatedly in the list. To prove the formula, use F to find one root, divide out its linear factor, and apply F again to each positive-degree quotient. Each step lowers the degree by one. After n steps the quotient is constant, equal to the original leading coefficient because all removed factors are monic, meaning leading coefficient 1. For $n = 0$, the product is empty and is defined to be 1. If a root α occurs k times in this list, collect those factors to get $P(z) = (z - \alpha)^k Q(z)$. The remaining factor satisfies $Q(\alpha) = a_n \prod_{\alpha_j \neq \alpha} (\alpha - \alpha_j) \neq 0$, since each of its finitely many factors is nonzero. The definition of multiplicity therefore identifies k as the multiplicity of α. Thus the root count with multiplicity is exactly n for $n \geq 1$.

For a real polynomial, conjugation gives

$$P(\overline{z}) = \sum_{j=0}^{n} a_j \overline{z}^{\,j} = \overline{\sum_{j=0}^{n} a_j z^j} = \overline{P(z)}.$$

Consequently, a nonreal root occurs together with its conjugate. Their multiplicities agree as well. To see this, suppose $P(z) = (z-\alpha)^k Q(z)$ with $Q(\alpha) \neq 0$. Let $Q^*(z)$ be the polynomial obtained by conjugating every coefficient of Q. Conjugating all coefficients of the polynomial identity gives

$$P(z) = (z-\overline{\alpha})^k Q^*(z), \qquad Q^*(\overline{\alpha}) = \overline{Q(\alpha)} \neq 0.$$

The left polynomial is unchanged because its coefficients are real. The nonzero final value proves that the conjugate root has exactly the same multiplicity k.

Theorem 16.3 (Real factorisation from F)**.** *Every nonzero real polynomial is a nonzero real constant times a product of real linear factors and real quadratic factors with no real roots. The factors may repeat.*

Proof. Use the complete complex factorisation just proved from F. Every real root supplies a real linear factor. Pair each nonreal root $\alpha = a + \mathrm{i}b$, with $b \neq 0$, with its conjugate, using their equal multiplicities. The paired factors multiply to

$$(z-\alpha)(z-\overline{\alpha}) = z^2 - 2az + (a^2+b^2).$$

This polynomial has real coefficients and is positive on the real axis, since its value at $x \in \mathbb{R}$ is $(x-a)^2 + b^2 > 0$. It therefore has no real root. These pairs and the real linear factors use all the complex factors. The leading coefficient is the required nonzero real constant. ∎

A real quadratic with no real root is *irreducible over* $\mathbb{R}$: it cannot be written as a product of two positive-degree real polynomials. Indeed, degrees would force two linear factors, each of which has a real root. The proof of Theorem 16.3 is a consequence of the explicit standing assumption F. It is not a proof of F itself.

Example 16.4 (Real factors and a repeated root)**.** Factor the real polynomial

$$P(x) = x^6 - 4x^5 + 4x^4 + 4x^2 - 16x + 16.$$

Grouping gives

$$P(x) = (x^2 - 4x + 4)(x^4 + 4) = (x-2)^2(x^4+4).$$

The remaining fourth-degree factor satisfies

$$x^4 + 4 = (x^2+2)^2 - (2x)^2 = (x^2 - 2x + 2)(x^2 + 2x + 2).$$

The two quadratics are $(x-1)^2 + 1$ and $(x+1)^2 + 1$, so they have no real roots. The full real factorisation is therefore

$$P(x) = (x-2)^2(x^2 - 2x + 2)(x^2 + 2x + 2).$$

Its only real root is 2, with multiplicity 2 because the remaining product has value $2^4 + 4 = 20 \neq 0$ there. The four nonreal roots are $1 + \mathrm{i}$, $1 - \mathrm{i}$, $-1 + \mathrm{i}$ and $-1 - \mathrm{i}$, each with multiplicity 1. Together with the repeated real root, they account for the degree 6.

16.4 Differentiation with a real parameter

Let $I \subset \mathbb{R}$ be an open interval. A complex-valued function $F : I \to \mathbb{C}$ has real component functions u and v with $F(t) = u(t) + \mathrm{i}v(t)$. A limit in $\mathbb{C}$ means that the modulus of its error tends to zero. This is equivalent to convergence of both real components, because

$$|\operatorname{Re} w| \leq |w|, \qquad |\operatorname{Im} w| \leq |w|, \qquad |w| \leq |\operatorname{Re} w| + |\operatorname{Im} w|.$$

The first two inequalities follow from the squared modulus. The third is the triangle inequality for the real and imaginary parts.

When both components are differentiable at t, define the *derivative with respect to the real parameter* by

$$F'(t) = u'(t) + \mathrm{i}v'(t).$$

The component-limit equivalence shows that this is precisely the limit of $(F(t+h)-F(t))/h$ as the real number h tends to zero. Conversely, existence of that limit gives the two component derivatives. Higher derivatives mean successive derivatives with the same real parameter. In particular, $(\operatorname{Re} F)' = \operatorname{Re}(F')$ when the derivative exists.

Fix $\lambda = a + \mathrm{i}b$ and let $F(t) = e^{\lambda t}$ for $t \in \mathbb{R}$. The real expression for the complex exponential is $F(t) = e^{at}(\cos bt + \mathrm{i}\sin bt)$. The real product and chain rules give

$$F'(t) = e^{at}\big((a\cos bt - b\sin bt) + \mathrm{i}(a\sin bt + b\cos bt)\big) = \lambda e^{\lambda t}.$$

Multiplication by a constant complex number is a real linear combination of the two components, so it commutes with this differentiation. Induction therefore gives

$$\frac{\mathrm{d}^n}{\mathrm{d}t^n} e^{\lambda t} = \lambda^n e^{\lambda t} \qquad (n \geq 0). \tag{16.3}$$

For $n = 0$, the factor λ^0 means 1, including when $\lambda = 0$. This statement varies t along the real axis. It introduces no derivative with a complex increment.

16.5 Phasors and circular motion

Fix an angular frequency $\omega > 0$. A *phasor* is a complex constant A representing the real motion

$$x(t) = \operatorname{Re}(Ae^{\mathrm{i}\omega t}), \qquad t \in \mathbb{R}. \tag{16.4}$$

If $A = R\operatorname{cis}\varphi \neq 0$, then $x(t) = R\cos(\omega t + \varphi)$. The *amplitude* is $R = |A| > 0$ and a *phase* is any argument φ of A, understood modulo 2π. For $A = 0$, the motion is zero, its amplitude is zero and there is no argument from which to assign a phase. For nonzero A, the least positive period is $2\pi/\omega$. Indeed, a period $T > 0$ evaluated at $t = -\varphi/\omega$ must satisfy $\cos(\omega T) = 1$, hence ωT is a positive integer multiple of 2π. The displayed least value is a period by the trigonometric identity. The zero motion has every positive period and no least positive one.

For real C and D, the expression $C\cos\omega t + D\sin\omega t$ has phasor $A = C - \mathrm{i}D$. Distribution verifies this sign: its real part is $C\cos\omega t + D\sin\omega t$. Adding motions at the same frequency adds their phasors, since real parts add. Cancellation of the phasor gives zero motion. For example, $3\cos 2t + 4\sin 2t$ has phasor $3 - 4\mathrm{i}$, amplitude 5 and phase $-\arctan(4/3)$. Adding $-\cos 2t + 2\sin 2t$, whose phasor is $-1 - 2\mathrm{i}$, gives phasor $2 - 6\mathrm{i}$, amplitude $2\sqrt{10}$ and phase $-\arctan 3$. The resulting motion is $2\cos 2t + 6\sin 2t$. Each phase listed is an argument; adding a whole turn gives another valid phase. Formula (16.3) also verifies $x'' + \omega^2 x = 0$ for every such motion, as in Lecture 11. Completeness of solution families is proved in Lecture 34.

For a circle of radius $R > 0$, the complex position

$$q(t) = Re^{\mathrm{i}(\omega t + \varphi)}$$

represents the planar position vector $(\operatorname{Re} q(t), \operatorname{Im} q(t))$. Define velocity as its componentwise derivative $v = q'$ and acceleration as $a = q''$. The derivative formula gives

$$v(t) = \mathrm{i}\omega q(t), \qquad a(t) = -\omega^2 q(t).$$

Multiplication by i is the positive quarter-turn, so velocity is tangent to the circle. The negative real factor makes acceleration point towards the centre. The speed is $|v| = R\omega$, the acceleration magnitude is $|a| = R\omega^2$, and the least positive period is $2\pi/\omega$. These are consequences of the prescribed path, not a force law. Figure 16.1 shows the path and derivative directions in the example.

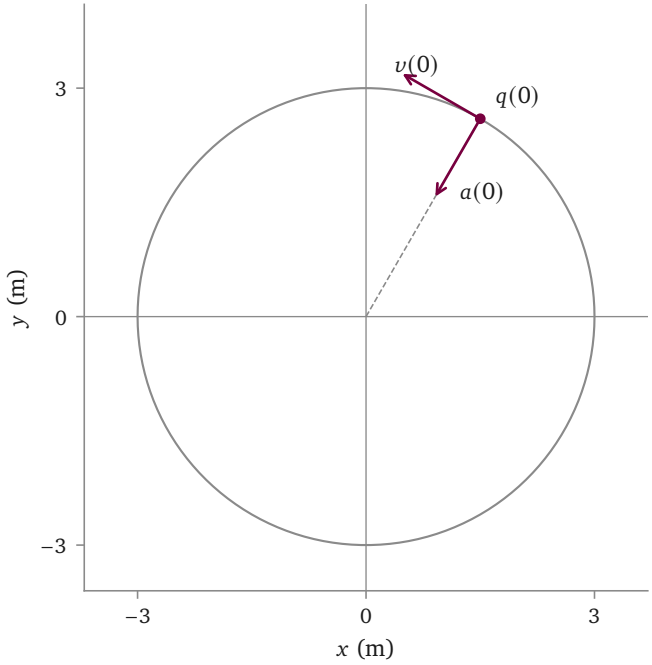


Figure 16.1: The path $q(t) = 3e^{\mathrm{i}(2t+\pi/3)}$ over one period. At $t = 0$, velocity is tangent and acceleration points towards the origin. The arrows show directions with a common drawing length, not physical magnitudes.

Example 16.5 (Position and its two derivatives)**.** Take $q(t) = 3e^{\mathrm{i}(2t+\pi/3)}$ metres with t in seconds. Its radius is 3 metres and its angular frequency is 2 radians per second. At $t = 0$, the position and derivatives are

$$q(0) = \left(\frac{3}{2} + \frac{3\sqrt{3}}{2}\mathrm{i}\right)\mathrm{m}, \qquad v(0) = (-3\sqrt{3} + 3\mathrm{i})\ \mathrm{m/s},$$

$$a(0) = (-6 - 6\sqrt{3}\,\mathrm{i})\ \mathrm{m/s^2}.$$

The speed is 6 metres per second, acceleration magnitude is 12 metres per second squared, and period is π seconds. Lecture 32 extends this componentwise method to general paths.

Summary 16.

- Equation (16.1): the complex exponential is defined by real functions. It obeys the product law, never vanishes and has imaginary period $2\pi\mathrm{i}$. Equation (16.2) retains all argument representatives.
- Theorem 16.1: polynomial division has a unique lower-degree remainder. Linear division gives the factor theorem and root multiplicity.
- Standing fact F supplies a root to every nonconstant complex polynomial. Theorem 16.3 proves the resulting real linear/quadratic factorisation by pairing conjugates with equal multiplicity.
- Equation (16.3): differentiation uses a real parameter. Phasors give sinusoidal amplitudes and phases; the zero motion has no assigned argument or least positive period.
- A prescribed circle with $R > 0$ and $\omega > 0$ has tangential velocity $\mathrm{i}\omega q$ and inward acceleration $-\omega^2 q$.

Sheet 16 develops exponential equations, real factors and phasor calculations. Lecture 17 begins the systematic solution of linear equations.

References

K. F. Riley, M. P. Hobson and S. J. Bence, *Mathematical Methods for Physics and Engineering*, 3rd ed., Cambridge University Press, 2006, Chapter 3.